Discover Artificial Intelligence

RESEARCH Open Access

An efficient deep learning model for reliable detection and classification of retinopathy of prematurity

Esraa Mohiy[1], Hamada I. AbdulWakel[2], Marwa Khairy[2] and Essam H. Houssein[2,3*]

*Correspondence:
Essam H. Houssein
essam.halim@mu.edu.eg
[1]Computer Science Department, Deraya University, Minia, Egypt
[2]Faculty of Computers and Information, Minia University, Minia, Egypt
[3]Minia National University, Minia, Egypt

Abstract

Retinopathy of prematurity (ROP) is a leading cause of childhood blindness in very low birthweight and premature infants, and its timely diagnosis is hampered by the shortage of specialized clinicians. Recent advances in artificial intelligence have demonstrated expert-level performance in detecting severe ROP in retinal images, motivating the development of automated screening tools. This work introduces ROPDeepX, a hybrid convolution neural network (CNN) architecture that integrates ResNet50 and EfficientNet-B4 architectures to enhance ROP detection and severity classification. To address the scarcity of publicly accessible ROP datasets, three datasets were employed: the newly released FARFUM-RoP dataset, the Retinal Image Dataset of Infants and ROP (RIDIRP), and the HVDROPDB dataset.The FARFUM-RoP dataset served as the primary source for model training, while RIDIRP and HVDROPDB were used to validate cross-dataset robustness. VGG19, ResNet50, EfficientNet-B4, and ROPDeepX were evaluated on these datasets. ROPDeepX outperformed the pretrained models, achieving 96. 9% accuracy on the FARFUM-RoP dataset. ROPDeepX demonstrated superior results in different evaluation metrics, in both multiclass and binary classification tasks.

Keywords Retinopathy of prematurity (ROP), Convolutional neural networks (CNN), Plus disease classification, ResNet50, EfficientNet-B4, VGG19

1 Introduction

ROP is affecting approximately 80% of premature infants with birth weight less than 1000 g[1]. ROP is an ailment related to the underdeveloped retina and blood vessels in preterm infants [2]. ROP may be present, within certain guidelines, for all premature infants having $\leq$ 1500g birth weight or a gestational age of 30 weeks or less as per the attending neonatologist's decision, and sometimes even infants with a birth weight ranging from 1500 g to 2000 g or greater than 30 weeks of gestation [3]. Advancements in neonatal care are increasing the rate of premature delivery and survival; however, the number of infants at risk for ROP has been rising globally, especially in middle-income nations [4].

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

ROP is marked by pathological neovascularization in the immature retina. Without timely intervention, this disorganized neovascularization can trigger fibrous tissue development, cause traction on the retina, and ultimately lead to retinal detachment, resulting in permanent vision loss [5]. Despite advances in neonatal care, its early and accurate detection remains challenging due to the continued dependence on manual retinal image screening by ophthalmologists.This process is time-consuming and prone to variability between clinicians, making automated solutions essential for improving early diagnosis and treatment outcomes.ROP is described in many ways: three zones, five stages, and a plus disease of the retina. This study focused on plus and pre-plus disease classification.This classification focuses on vascular changes rather than the overall stage of ROP. It assesses the severity of blood vessel dilation and tortuosity (twisting)[6], which can indicate a more aggressive disease process.

In Fig. 1 (a) represents a diagnosis of a normal retina, (b) illustrates pre-plus disease, characterized by noticeable but not severe tortuosity of the posterior pole vessels, particularly visible in the third quadrants, insufficient for a plus disease diagnosis, and (c) depicts plus disease, where significant vascular tortuosity and dilation of the posterior pole vessels are evident in the first and fourth quadrants [7]. The red arrows highlight these abnormal vascular changes.

Deep learning, particularly with the implementation of Convolutional Neural Networks (CNNs), has proven to be highly effective in the automated analysis of medical imaging, particularly in the detection and classification of retinal pathology. CNNs are adept at feature extraction from complex images, making them ideal for diagnosing conditions such as ROP[8]. However, there is still potential to improve model accuracy, robustness, and generalization across different datasets and disease stages.

1.1 Motivation

ROP remains a significant cause of preventable childhood blindness, particularly among preterm and low-birthweight infants. Accurate and timely diagnosis is essential for preventing disease progression, yet current screening practices rely heavily on manual interpretation by ophthalmologists, which is both resource-intensive and prone to inter-observer variability.

Deep learning–based approaches have shown promising results in automated ROP detection and classification; however, most prior studies have been conducted on non-public or institution-specific datasets, limiting reproducibility, transparency, and the ability to compare results across research groups. Moreover, the scarcity of openly accessible, high-quality ROP datasets has hindered progress toward robust and generalizable models, especially for challenging distinctions such as between Pre-Plus and Plus disease.

To address these gaps, this study uses only publicly available ROP datasets, including the newly released FARFUM-RoP, employed here in one of the early studies developing

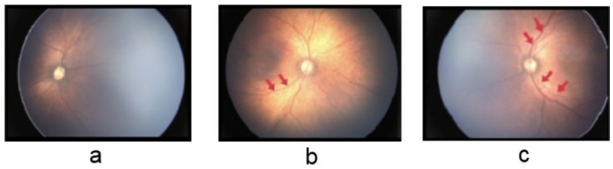


Fig. 1 Plus and pre-plus disease classification

and evaluating an automated ROP classification model. By using exclusively open-access datasets, the work ensures reproducibility, facilitates fair benchmarking, and promotes broader adoption of AI solutions in neonatal ophthalmology.

1.2 Contribution

This study introduced ROPDeepX, a hybrid deep learning model developed using the FARFUM-RoP[10] dataset, which served as the foundation for model training. ROPDeepX combines the strengths of ResNet50 and EfficientNet-B4 architectures to deliver enhanced performance in the classification of retinal images. After building and training the model on FARFUM-RoP, The performance was evaluated on several additional datasets (RIDIRP[11] and HVDROPDB [12]), assessing its ability to classify retinal images into three categories: Normal, Pre-Plus, and Plus, as well as its capability for binary classification (Normal vs. ROP), as described in Table 1.

The main contributions of this study are as follows:

1. Novel Hybrid Architecture (ROPDeepX):Introduction of ROPDeepX, a hybrid deep learning model that integrates ResNet50 and EfficientNet-B4 backbones with a soft attention–based feature fusion mechanism to enhance ROP detection and classification.
2. Exclusive Use of Open-Access Datasets: Unlike most prior studies that relied on non-public datasets, ROPDeepX is trained and evaluated exclusively on publicly available ROP datasets, namely FARFUM-RoP, RIDIRP, and HVDROPDB. Notably, FARFUM-RoP is used here as part of the early body of studies developing and evaluating automated ROP classification models.
3. Evaluation Across Multiple Classification Tasks: Model performance is assessed in both multi-class (Normal, Pre-Plus, Plus) and binary (Normal vs. ROP) classification settings on all datasets, demonstrating robustness across diverse imaging sources and classification complexities.
4. Superior Performance Over Baselines: Comparative experiments indicate that ROPDeepX consistently outperforms established CNN architectures (VGG19, ResNet50, EfficientNet-B4) across all evaluation metrics, achieving up to 99.7% accuracy in multiclass classification and 99.8% AUC in binary classification
5. Enhanced Generalization Capability: The proposed model demonstrates strong cross-dataset generalization, accurately distinguishing between closely related disease stages–a key challenge in clinical ROP screening.
6. Comprehensive Evaluation Metrics: The review process employs multiple performance indicators, including accuracy, sensitivity, specificity, precision, F1-score, and AUC, ensuring a thorough assessment of model effectiveness.

Table 1 Stage of plus disease classification

Term	Description
Normal	Regular, straight vessels without any abnormal dilation or tortuosity
Pre-plus	Mild dilation and tortuosity of the vessels, but not enough to be classified as Plus Disease. It indicates early stages of worsening disease, which requires careful monitoring and early intervention if needed
Plus disease	Severe dilation and tortuosity of the retinal vessels in at least 2 quadrants of the posterior retina. This indicates aggressive disease and usually requires immediate treatment such as laser therapy or anti-VEGF injections [9] to prevent retinal detachment or vision loss

1.3 Paper structure

The remaining sections are arranged as follows

- Section 2—Literature Review: Reviews previous studies on automated detection and classification of ROP, highlighting recent advancements, existing limitations, and research gaps addressed in this work.
- Section 3—Methodology: Describes the datasets, preprocessing techniques, proposed ROPDeepX architecture, training procedure, and evaluation metrics used in this study.
- Section 4—Results and Discussion: Presents the experimental findings, including performance comparisons between ROPDeepX and baseline models across multiple datasets and classification tasks, followed by an interpretation of the results and their implications for automated ROP screening. Discusses the study's limitations and outlines possible directions for further research.
- Section 5—Conclusion: Summarizes the key contributions and findings of the study, highlighting the significance of ROPDeepX in improving ROP detection and classification.

2 Literature review

Deep learning has become an increasingly prominent approach in automating the detection and classification of ROP, particularly for identifying plus disease and staging severity.

Early CNN-based approaches achieved promising results on specific datasets but often lacked generalizability. For instance, Agrawal et al. [13] employed a CNN–LSTM framework with architectures such as EfficientNet-B4, ResNet152, and Xception, achieving 93% accuracy on cropped STAGE and HVDROPDB-PLUS datasets. Huang et al. [14] enhanced ConvNeXt with adaptive attention modules, obtaining an AUC of 89.4% and accuracy of 81.8% on a clinical dataset. Similarly, Salih et al.[15] compared VGG19, ResNet50, EfficientNetB5, and custom CNN models, reporting accuracy of 88.82% with balanced precision and recall values.

Hybrid and ensemble methods have been explored to improve robustness and interpretability. Ramanathan et al.[16]systematically reviewed AI algorithms for ROP diagnosis.The review covered hybrid, and advanced deep learning approaches and RetCam images datasets, obtaining accuracy in the range of 85–98%. Sankari and Umapathy [17] applied multiple CNN backbones, including ResNet50, ShuffleNet V2, EfficientNet, and VGG16, reaching 89.5% accuracy with an AUC of 92% on the RETINOPRETERM dataset. Mutua et al. [18] integrated ResNet50 with deep neural networks (DNN) for combined Kaggle and HVDROPDB datasets, achieving 92.8% accuracy and high sensitivity 94.9%.

Segmentation-focused approaches have also contributed to ROP analysis. Matamala Avrova et al. [19] used U-Net architecture for plus disease segmentation within the RTOC – Catalonia Telemedicine Network, achieving 81.04% accuracy. Agrawal et al. [13] employed U-Net and ResNet50 with a shape-aware module (SAM) and Circle Hough Transform to segment vessels, yielding sensitivity rates of 95.72% for ROP and 91.13% for plus disease. Sankari et al.[17] applied MultiResUNet to a private dataset of 4,000 images, obtaining 94.5% accuracy with high sensitivity and specificity values.

Large-scale supervised learning studies have also been conducted. Tong et al. [20] used a 101-layer ResNet to analyze over 36,000 fundus images from multiple experts, reporting 90.3% accuracy in detecting both ROP severity and plus disease. Huang et al. [14] evaluated VGG19, VGG16, InceptionV3, MobileNet, and DenseNet on a local dataset, with VGG19 achieving the highest accuracy 96% and specificity 95.2%. Chen et al.[21] demonstrated cross-population generalizability by training CNNs on datasets from North America and Nepal, achieving sensitivity of 94%.

Semi-supervised and specialized detection frameworks have also emerged. Feng et al. [22]developed a semi-supervised learning framework on 4,796 images from Xinhua Hospital, reaching 86.3% AUC. Attallah [23] introduced DiaROP, an automated diagnostic tool combining VGG, MobileNet, Inception, and DenseNet, achieving 93.2% accuracy and 98% AUC. Mulay et al. [24]applied an R-CNN for early-stage detection on KIDROP images, achieving 88% accuracy. Tan et al. [25]used CNNs on 6,974 Australasian fundus images, obtaining 81.4% sensitivity and 80.7% specificity.

While these works demonstrate substantial progress, most rely on non-public or proprietary datasets as illustrated in Table 2, limiting reproducibility and cross-study comparisons. Performance often drops when models are tested on datasets from different institutions or imaging devices, indicating challenges in generalization. Furthermore, differentiating between clinically similar stages such as Pre-Plus and Plus remains a persistent difficulty, even for advanced models.

Table 2 Summary of publicly reported ROP datasets and their availability

Dataset	Images	Labels/tasks	Availability
FARFUM–RoP (Iran)[10]	1 533	Normal,Pre–Plus, and Plus	Public via Figshare
HVDROPDB (India) [12]	600* (50 per sub–dataset)	Optic disc, vessels and ridge segmentation	Public via Mendeley Data
RIDIRP / Czech–Ostrava [11]	6 004	Normal and 14 other diagnoses; Normal, Pre-Plus, and Plus	Free
USA-iROP [26, 27]	5 561	Normal, Plus, and Pre-Plus	Not publicly available
USA–1 [28]	77	Normal, Plus, and Pre-Plus	Not publicly available
Canada (Worrall) [29]	1 459	Plus and No Plus (per exam and per eye)	Not publicly available
UK–London [29]	106	Plus and No Plus	Not publicly available
USA–2 part of iROP [30]	5 512	Normal, Plus, and Pre–Plus	Not publicly available
USA–3 [31]	2759	Stage1, Stage2, and Stage3	Not publicly available
Germany [32]	–	–	Available on demand; images not published
South Korea [33]	–	–	Available on demand; images not published
China – Chengdu [34]	20 795	Normal / Minor ROP / Severe ROP	Not publicly available
USA–4 (Chiang et al.) [35]	248 eyes	–	Not publicly available
USA–5 [30, 36]	100	Normal / Plus / Pre–Plus	Not publicly available
China – Wuchan [20]	38 895	Normal / Mild / Semi–Urgent / Urgent; ridge/detachment identification	Available on demand
China – Shenzhen [37]	20 822	Disease / Normal / Unqualified	Original data not available
Taiwan & Japan [38]	6 500	Normal / Stage 1–3	Available on demand
India (Bhatkalkar et al.) [39]	6 043	–	Not publicly available
USA–SUNDROP [40, 41]	79 807	No / Mild ROP / MTM–ROP / Treatment–warranted ROP	Not publicly available
ROP[42, 43]	4 113 cases	Stage 1–4 (demarcation line, ridge, extra-retinal neovascularization, detachment)	Not publicly available

* HVDROPDB comprises three sub-datasets, each with 50 images and corresponding masks

To address these challenges, the present study introduces ROPDeepX, a hybrid CNN architecture combining ResNet50 and EfficientNet-B4 with a soft attention–based feature fusion mechanism. Unlike most prior works, ROPDeepX is trained and evaluated exclusively on publicly available datasets, including the newly released FARFUM-RoP, and is assessed on both binary and multiclass classification tasks to ensure robustness, reproducibility, and cross-dataset generalization.

3 Methodology

The proposed research introduces ROPDeepX, a hybrid deep learning framework specifically designed for the automated detection and severity grading of ROP from retinal fundus imagery. The methodological pipeline comprises three integral phases: (i) acquisition and preprocessing of high-quality annotated datasets, (ii) design and implementation of a novel hybrid architecture that integrates ResNet50 and EfficientNet-B4 backbones through a soft attention–based feature fusion mechanism, and (iii) rigorous performance evaluation using multiple complementary datasets and clinically relevant metrics. This structured approach ensures both technical robustness and clinical applicability, while promoting reproducibility through the exclusive use of publicly accessible datasets.

3.1 Dataset

ROPDeepX was trained and evaluated using three publicly available datasets: FARFUM-RoP [10], RIDIRP [11], and HVDROPDB [12]. Notably, This study utilizes the FARFUM-RoP dataset, a newly developed collection of high-resolution retinal fundus images annotated for ROP stages. To ensure comprehensive evaluation and to assess the model's generalization ability across varied imaging conditions, the RIDIRP and HVDROPDB datasets were also employed.

- *FARFUM-RoP*[10] created through a collaboration between Farabi Eye Hospital and Ferdowsi University of Mashhad, introduces a novel high-quality source of images that broadens the scope of reproducible research in ROP detection. It includes 1,533 fundus images, was captured using the RetCam Shuttle wide-field from 68 patients that have been independently labeled "Normal," "Pre-Plus," or "Plus" by five experienced pediatric ophthalmologists. This dataset served as the foundation for training and testing the hybrid deep learning model.
- *Retinal Image Dataset of Infants and ROP (RIDIRP)*[11], which contains 6,004 retinal images of premature infants, was acquired using Clarity RetCam 3, Natus RetCam Envision, and Phoenix ICON with a focus on multiclass classification tasks. The dataset is accompanied by the anonymized patients' information from the ROP screening acquired at the University Hospital Ostrava, Czech Republic. Three experts independently classified the images with very comparable results to the plus form and the stages form.
- *HVDROPDB V3*[12] dataset is a publicly accessible dataset of retinal images utilized for segmentation and Normal vs. ROP classification. It includes 100 retinal images, which were captured using RetCam Shuttle/RetCam 3 and 3nethra Neo, chosen from temporal and posterior views. It was given its name in honor of the H.V. Desai Eye Hospital in Pune, India, which gathered fundus images of premature babies.

For binary classification tasks, the Pre-Plus and Plus disease categories in the FARFUM-RoP and RIDIRP datasets were merged into a single "ROP" category, resulting in two classes: Normal and ROP. This consolidation was motivated by both clinical and methodological considerations. Clinically, Pre-Plus and Plus represent progressive stages along the same pathological continuum of ROP, where distinguishing between them is less critical for initial screening than differentiating between healthy and diseased eyes. Methodologically, merging these categories reduces class imbalance, increases the number of positive samples, and simplifies the classification task, thereby allowing the model to focus on the broader goal of detecting any presence of ROP. This process generated two additional datasets–FARFUM-RoP (2-Classes) and RIDIRP (2-Classes)–for evaluating the generalizability and robustness of ROPDeepX across both multiclass and binary classification settings.Table 3 Summary of the datasets utilized in this study, including class distribution, and their role in developing and evaluating the ROPDeepX model.

3.2 Data preprocessing

The preprocessing and augmentation pipeline was designed to prepare retinal fundus images for subsequent analysis. Figure 2 outlines the preprocessing pipeline applied to the retinal fundus dataset. The images were split with a three-fold cross-validation (70 % training, 15 % validation and 15 % testing.) Training–validation images underwent random resized cropping (380 × 380 px), horizontal/vertical flips, rotations within ±15°, and color jitter adjustments to brightness, contrast, and saturation, followed by conversion into multi-dimensional numerical arrays and per-channel normalization. Validation and test images were resized to 400 px, center-cropped to 380 × 380 px, converted into multi-dimensional numerical arrays, and normalized. All data were loaded in batches of eight, producing standardized inputs ready for model development. The data loading mechanism facilitates efficient mini-batch processing and leverages hardware acceleration on either GPU or CPU.The output of this pipeline is a set of uniformly preprocessed and augmented images, ready to be fed into the ROPDeepX hybrid deep learning model for classification of ROP.

This approach introduces controlled variability, ensuring that the dataset better reflects real-world imaging conditions while minimizing overfitting risks.

3.3 Hybrid model architecture

The proposed ROPDeepX architecture integrates two complementary convolutional neural network (CNN) backbones–ResNet-50 and EfficientNet-B4–to leverage the strengths of both residual learning and compound scaling. Figure 3 illustrates the

Table 3 Datasets used in training and evaluation ROPDeepX

Dataset Name	Images	Classes	Labeling experts	Purpose in study
FARFUM-RoP	1,533	Normal, Pre-Plus, Plus	5 pediatric ophthalmologists	Training and testing ROPDeepX (3-class)
RIDIRP	6,004	Normal, Pre-Plus, Plus	3 ROP experts	Generalization testing (3-class)
HVDROPDB V3	100	Normal, ROP	Not specified	Binary classification performance
FARFUM-RoP (2-Classes)	1,533	Normal, ROP	5 pediatric ophthalmologists	Training and testing ROPDeepX (2-class)
RIDIRP (2-Classes)	6,004	Normal, ROP	3 ROP experts	Binary classification performance

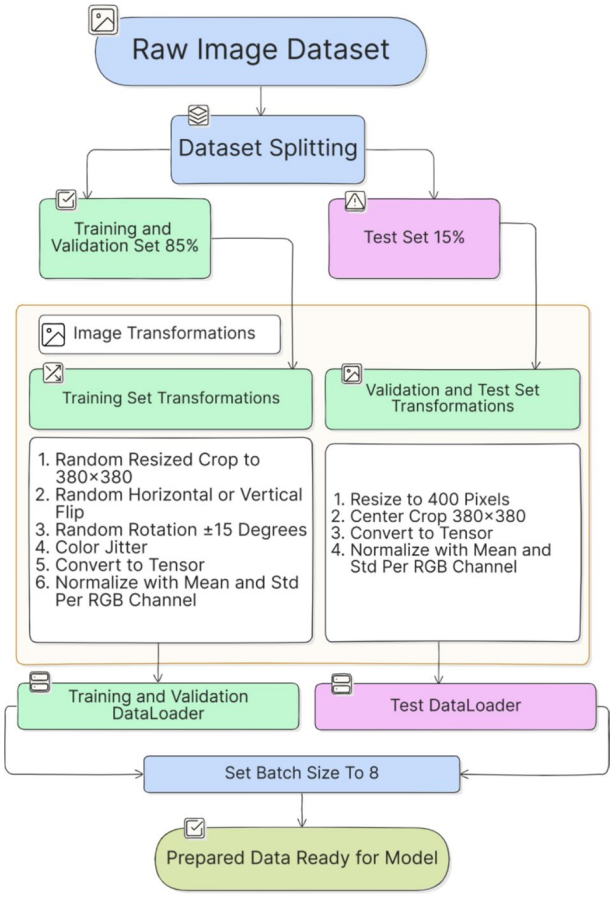


Fig. 2 ROPDeepX data preprocessing flow

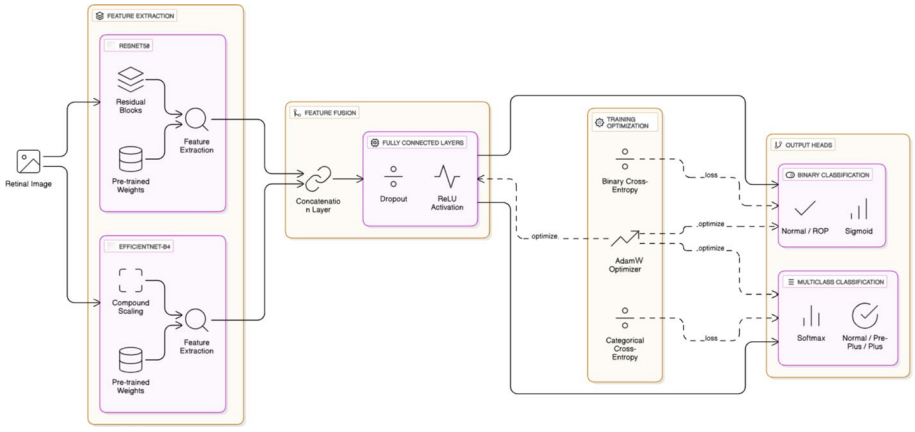


Fig. 3 ROPDeepX Model Architecture and Training Procedure

architecture and training flow, including the direction of data movement (solid arrows) and gradient propagation (dashed arrows).

3.3.1 Dual-branch feature extraction

The preprocessed retinal fundus image is simultaneously fed into two independent feature extraction branches:

- *Residual learning in ResNet-50* ResNet-50 is based on deep residual learning, where instead of learning a direct mapping, the network learns a residual function. Given an input feature map x_l, a residual block computes the output as:

$$x_{l+1} = x_l + \mathcal{F}(x_l, W_l) \tag{1}$$

- where x_l is the input to the l-th layer, $\mathcal{F}(x_l, W_l)$ is the residual function (typically composed of two or three convolutional layers with weights W_l), and the addition represents a skip connection. This formulation helps prevent vanishing gradients and allows training of deeper networks by preserving identity mappings across layers [44].*EfficientNet-B4 scaling principle* EfficientNet-B4 applies a compound scaling method to balance network depth, width, and input resolution. The scaling is governed by:

$$d = \alpha^{\phi}, \quad w = \beta^{\phi}, \quad r = \gamma^{\phi}, \quad \text{subject to} \quad \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \tag{2}$$

- where the model depth d, width w, and input resolution r are scaled using constants $\alpha, \beta, \gamma > 1$, determined via grid search. The user-defined scaling coefficient ϕ controls the overall resource budget. EfficientNet-B4 uses these parameters to scale the base model EfficientNet-B0 to a deeper and wider network, while keeping computational efficiency optimized [45].ResNet-50 outputs a 2048-dimensional feature vector capturing high-level discriminative features such as vessel branching patterns and localized lesions.
- EfficientNet-B4 generates a 1792-dimensional vector that encodes fine-grained texture and global retinal structures.

3.3.2 Soft attention-based feature fusion

To effectively integrate complementary representations from the ResNet-50 and EfficientNet-B4 branches, we employ a soft attention-based feature fusion mechanism. Let $\mathrm{f}_r \in \mathbb{R}^d$ denote the projected feature vector obtained from the ResNet-50 branch after dimensionality reduction, and $\mathrm{f}_e \in \mathbb{R}^d$ represent the corresponding projected feature vector from the EfficientNet-B4 branch. These two vectors are concatenated to form:

$$\mathrm{f}_c = [\mathrm{f}_r \,\|\, \mathrm{f}_e] \in \mathbb{R}^{2d}, \tag{3}$$

where $\|$ denotes the concatenation operation.

The concatenated vector f_c is then passed through a fully connected (FC) layer, followed by batch normalization (BN), ReLU activation, and dropout:

$$\mathrm{f}_j = \mathrm{Dropout}(\sigma_{\mathrm{ReLU}}(\mathrm{BN}(W_f \mathrm{f}_c + b_f))), \tag{4}$$

where $W_f \in \mathbb{R}^{d \times 2d}$ and $b_f \in \mathbb{R}^d$ are learnable parameters.

An attention vector $\mathrm{a} \in \mathbb{R}^2$ is then computed by a second FC layer followed by the softmax function:

$$\mathrm{a} = \mathrm{Softmax}(W_a \mathrm{f}_j + b_a), \tag{5}$$

where $W_a \in \mathbb{R}^{2 \times d}$ and $b_a \in \mathbb{R}^2$ are learnable parameters. The softmax ensures that $a_1 + a_2 = 1$ and $a_i \geq 0$ for $i \in \{1, 2\}$, thereby producing non-negative, normalized attention weights.

Finally, the weighted fused feature vector is calculated as:

$$\mathrm{f}_{\mathrm{att}} = a_1 \cdot \mathrm{f}_r + a_2 \cdot \mathrm{f}_e. \tag{6}$$

This mechanism enables the network to dynamically adjust the contribution of each branch for a given input, emphasizing the more informative features for robust classification.

3.3.3 Projection and classification heads

To ensure compatibility between the heterogeneous feature representations produced by the two backbones (ResNet-50 and EfficientNet-B4), each branch is followed by a *projection head* that reduces and aligns their dimensionality. Specifically, the ResNet-50 backbone outputs a feature vector of size 2048, while the EfficientNet-B4 backbone outputs a vector of size 1792. Each projection head consists of a fully connected (FC) layer that maps the respective feature vector to a shared dimensionality of 1024:

$$\mathrm{z}_r = \mathrm{W}_r \mathrm{f}_r + \mathrm{b}_r, \quad \mathrm{z}_e = \mathrm{W}_e \mathrm{f}_e + \mathrm{b}_e, \tag{7}$$

where f_r and f_e are the raw backbone features, z_r and z_e are the projected features, and $\mathrm{W}_r, \mathrm{W}_e$ are learnable weight matrices.

These projected features are then processed by the *soft attention-based feature fusion* mechanism (Sect. 3.3.2), which produces an attended feature vector $\mathrm{f}_{att} \in \mathbb{R}^{1024}$. This attended vector serves as the input to the *classification head*.

The classification head is implemented as a fully connected layer mapping the 1024-dimensional attended feature vector to the number of target classes *C*:

$$\mathrm{o} = \mathrm{W}_c \mathrm{f}_{att} + \mathrm{b}_c, \tag{8}$$

where $\mathrm{o} \in \mathbb{R}^C$ are the class logits. A softmax activation function is subsequently applied to o to obtain the class probability distribution:

$$p(y = c \,|\, \mathrm{x}) = \frac{\exp(o_c)}{\sum_{j=1}^{C} \exp(o_j)}, \quad c \in \{1, \ldots, C\}. \tag{9}$$

In summary, Features from EfficientNet-B4 and ResNet-50 are projected to 1024 dimensions, concatenated, and refined through a fusion block. A soft attention mechanism assigns weights to each backbone's features, producing a weighted sum that is fed into a fully connected layer to output probabilities for Normal, Pre-Plus, and Plus classes (Fig. 4).

This architectural design ensures that (i) both backbone networks contribute equally in a compatible latent space, and (ii) the final prediction is derived from a learned combination of their complementary representations.

3.3.4 Training flow and backpropagation

The training procedure of the proposed hybrid model follows a standard forward–backward optimization loop with mini-batch processing. The process begins with the loading of preprocessed and augmented retinal fundus images in batches. Each image is simultaneously passed through the ResNet-50 and EfficientNet-B4 backbones, producing high-dimensional feature vectors. These features are projected into a shared latent space,

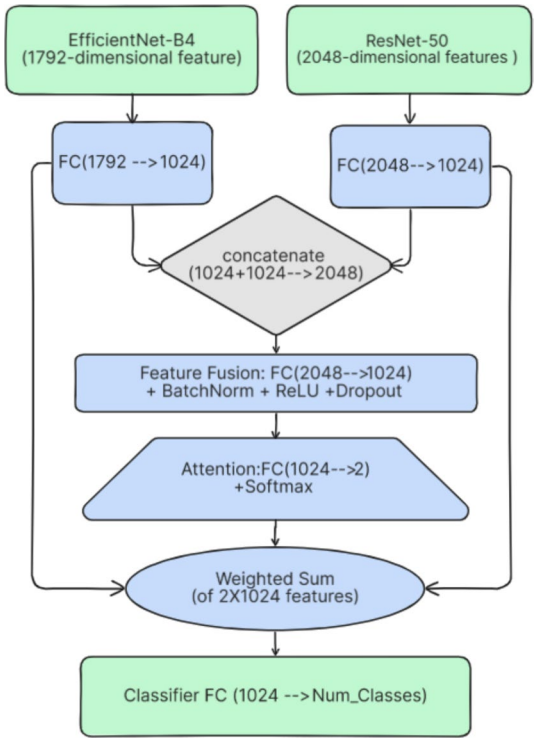


Fig. 4 Projection and classification head of the ROPDeepX model

fused using the soft attention mechanism, and then passed through the classification head to generate class predictions.

The predicted class probabilities are compared with the ground-truth labels using a cross-entropy loss function with label smoothing, which reduces overconfidence and improves generalization. The calculated loss is then propagated backward through the network using backpropagation, allowing gradients to update the trainable parameters of the projection heads, fusion block, attention mechanism, and selected layers of the backbone networks.

An AdamW optimizer is employed to perform parameter updates, while a OneCycleLR learning rate scheduler dynamically adjusts the learning rate throughout training. Mixed precision training is enabled to optimize memory usage and computational efficiency, making it possible to train with larger batch sizes and faster throughput. After each training epoch, the model is evaluated on a validation set to monitor performance and prevent overfitting.

This forward and backward pass sequence is repeated for a fixed number of epochs, gradually refining the model parameters until the optimal configuration is reached for accurate multiclass ROP classification.

3.4 Evaluation producer

The evaluation phase begins with model selection, where either the proposed ROP-DeepX hybrid model or a chosen pre-trained CNN architecture (VGG19, ResNet-50, or EfficientNet-B4) is specified. The selected model is then configured for two classification settings: multiclass classification and binary classification.

In the multiclass setting, experiments are conducted on the FARFUM-RoP and RIDIRP datasets, each containing three classes (Normal, Pre-Plus, Plus). For binary classification, the Pre-Plus and Plus categories are merged into a single ROP class, producing the FARFUM-RoP (2-Classes) and RIDIRP (2-Classes) datasets, alongside the HVROPDB V3 binary dataset.

On the FARFUM-RoP a three-fold cross-validation strategy was adopted to prevent patient overlap. For each fold, 70% of patients were used for training, 15% for validation, and 15% for testing then the results were reported as mean ± 95% bootstrap confidence intervals (CIs) across folds.

Performance metrics included per-class Precision, Recall, and F1-score (with supports), along with weighted accuracy, macro-averaged scores, and AUC.Normalized confusion matrices were generated to visualize prediction distributions across classes. To ensure well-calibrated probability outputs, calibration was evaluated on the test set using Expected Calibration Error (ECE), Brier Score, and Reliability Diagrams.Lower ECE and Brier values indicate better calibration.

Each model is trained and evaluated on the corresponding datasets under its classification setting. The results, including accuracy, precision, recall, F1-score, sensitivity, specificity, and AUC, are then compared across all models and datasets to assess relative performance and generalization capability.Figure 5 illustrates the steps of the evaluation

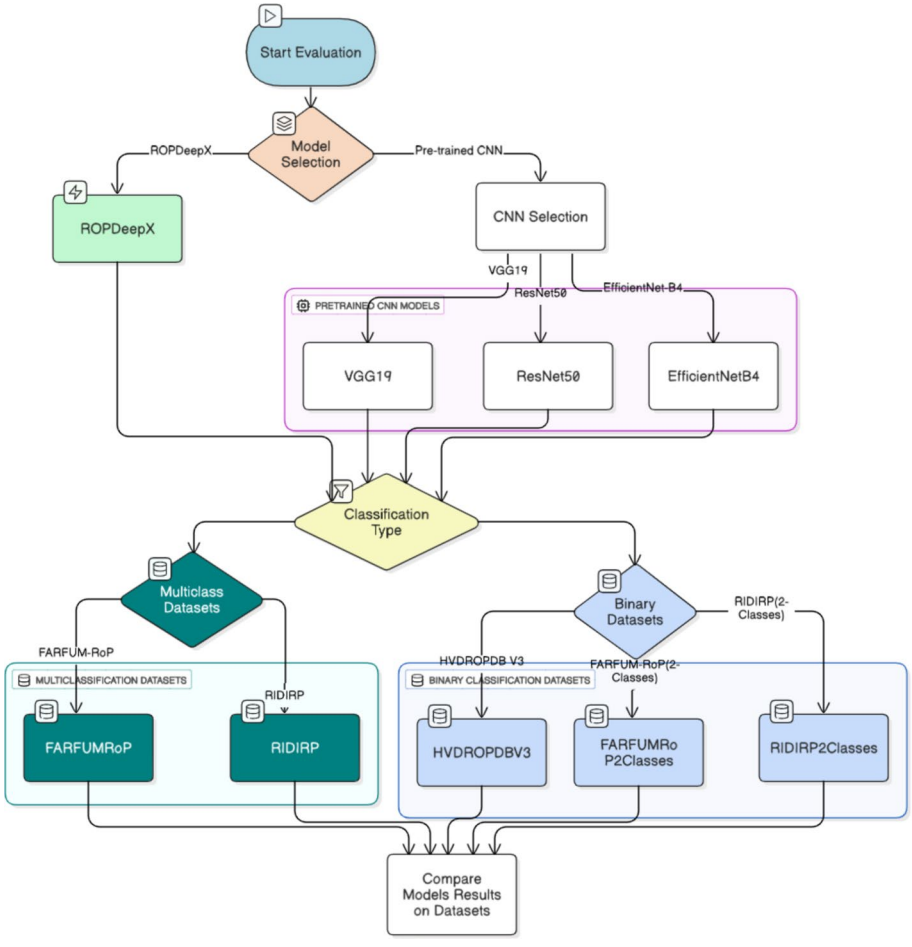


Fig. 5 Steps of evaluation ROPDeepX and selected CNN architectures

Table 4 Pre-class performance results for each class (Normal, Pre-Plus, Plus) across folds, showing Accuracy, Precision, Recall, F1, and Support. Support is the number of test samples in each fold for each class

Class	Accuracy (95% CI)	Precision (95% CI)	Recall (95% CI)	F1 (95% CI)	Support (Fold 1)	Support (Fold 2)	Support (Fold 3)
Normal	0.951 [0.935–0.965]	0.957 [0.940–0.973]	0.949 [0.930–0.967]	0.952 [0.935–0.969]	261	261	260
Pre-Plus	0.939 [0.919–0.960]	0.926 [0.910–0.942]	0.938 [0.922–0.954]	0.932 [0.920–0.945]	260	259	259
Plus	0.959 [0.945–0.974]	0.960 [0.945–0.974]	0.975 [0.960–0.988]	0.967 [0.954–0.980]	256	257	257
Mean ± 95% CI	**0.950±0.010**	**0.948±0.011**	**0.954±0.009**	**0.950±0.010**	–	–	–

Bold values indicates the best

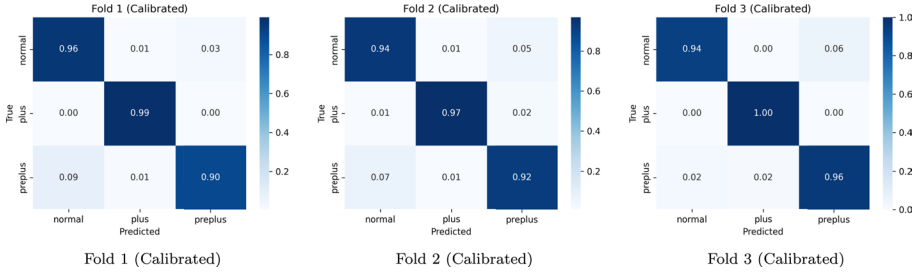


Fig. 6 Normalized confusion matrices across three folds: Fold 1, Fold 2, and Fold 3. These matrices display the true positive, false positive, and false negative rates for each class (Normal, Plus, Pre-Plus) after calibration

process, showing how each model was assessed on the different datasets and classification tasks.

4 Results and discussion

ROPDeepX model was trained and tested basically in FARFUM-RoP dataset using several performance metrics.This section presents the experimental results of the proposed ROPDeepX framework and compares its performance against baseline convolutional models across multiple datasets and evaluation settings.

4.1 Overall performance with three-fold cross-validation

Per-class performance (Table 4) demonstrates balanced detection across disease categories. The Plus class achieved the highest recall (97.5 %), while Pre-Plus showed slightly lower precision (92.6 %) owing to overlapping vascular features between stages. Figure 6 presents the normalized confusion matrices for each fold, confirming that most misclassifications occurred between Pre-Plus and Plus images, reflecting the inherent clinical difficulty of distinguishing borderline cases.

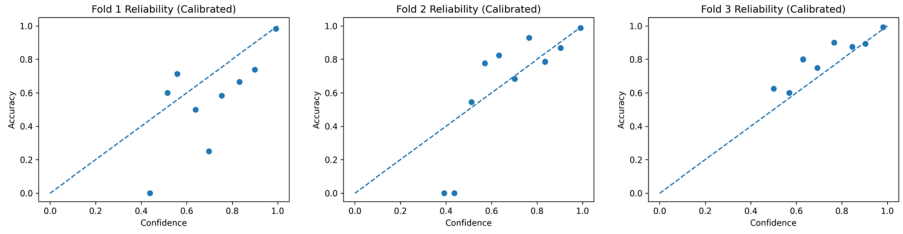
4.2 Calibration and Reliability

Table 5 presents the post-calibration performance of the **ROPDeepX** model across the three folds using Expected Calibration Error (ECE) and Brier Score as evaluation metrics. Lower values in both measures indicate better alignment between the model's predicted probabilities and the actual outcomes. The mean ECE of 0.0256 and Brier Score

Table 5 Calibration metrics (mean ± SD) across folds

Fold	ECE	Brier Score
1	0.0176	0.0318
2	0.0316	0.0335
3	0.0257	0.0381
Mean ± SD	**0.0256** ± **0.0071**	**0.0345** ± **0.0033**

Bold values indicates the best



(a) Fold 1 Reliability (Calibrated) (b) Fold 2 Reliability (Calibrated) (c) Fold 3 Reliability (Calibrated)

Fig. 7 Reliability diagrams for each fold, showing the relationship between predicted confidence and accuracy for the calibration process. These diagrams evaluate how well the model's predicted probabilities match the true accuracy across different confidence levels

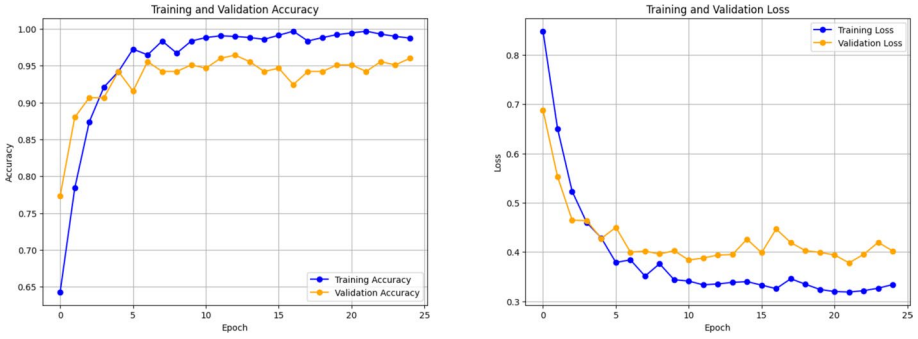


Fig. 8 Training and Validation Accuracy and Loss of ROPDeepX using FARFUM-RoP dataset for multiclass categories (Normal, Pre-Plus, Plus)

of 0.0345 show that the calibrated model produces well-calibrated confidence estimates with minimal over- or under-confidence. Reliability diagrams in Fig. 7 show post-calibration curves aligning closely with the ideal diagonal, indicating well-calibrated confidence estimates that enhance clinical interpretability. The small standard deviations across folds further confirm the stability and reliability of the calibration process.

The recall (sensitivity) values were also strong, with training at 99.69%, validation at 96.00%, and test at 97.36%, highlighting the model's effectiveness in detecting ROP cases. Precision was similarly impressive, with training at 99.45%, validation at 95.11%, and test at 97.04%, reflecting the model's ability to avoid false positives.

The AUC score further validated the model's performance, with values of 99.69%, 96.00%, and 99.32% for training, validation, and test, respectively, indicating strong discriminatory power.

Moreover,the training and validation curves for accuracy and loss in Fig. 8 demonstrate that ROPDeepX learns rapidly within the first few epochs, with accuracy exceeding 90% and loss dropping below 0.45 by epoch 5. Both metrics stabilize thereafter, with training accuracy approaching 100% and validation accuracy remaining above 95%,

while the loss curves show only a small, consistent gap, indicating minimal overfitting. These trends confirm the model's strong generalization and ability to maintain high performance across multi-class ROP classification tasks.

In general, ROPDeepX achieves state-of-the-art performance in detecting ROP, showing consistent accuracy, recall, precision, and specificity across FARFUM-RoP dataset. These results suggest that ROPDeepX could be a valuable model for early detection and timely treatment of ROP in clinical settings.

To further evaluate the performance of ROPDeepX, It was compared with three widely used pre-trained CNN architectures: VGG19, ResNet50, and EfficientNet-B4. Each model was trained on the FARFUM-RoP dataset, and their performance was measured using the same evaluation metrics (accuracy, precision, recall, specificity, F1-score, and AUC).The results clearly showed that ROPDeepX outperformed all three models in every metric.

4.3 Comparison of ROPDeepX with CNN architectures

To ensure a fair assessment of the proposed model's effectiveness, ROPDeepX was compared against three widely used CNN architectures – VGG19, ResNet50, and EfficientNet-B4 – using the same experimental configuration, including identical datasets(FARFUM-RoP), pre-processing steps, hyperparameters, and evaluation metrics. As shown in Fig. 9, ROPDeepX achieved the highest score in all metrics, with particularly notable gains in recall and F1-score, reflecting its improved ability to correctly identify ROP cases across different severity stages. Compared to VGG19, ROPDeepX improved recall by +17.42%, and compared to EfficientNet-B4, it achieved a +4.36% increase in AUC. These consistent improvements across multiple metrics highlight the advantages of the hybrid architecture and attention-based feature fusion mechanism, which enhance feature representation and discrimination. The results confirm that integrating complementary backbone networks enables more robust and accurate ROP classification than using any single CNN architecture.

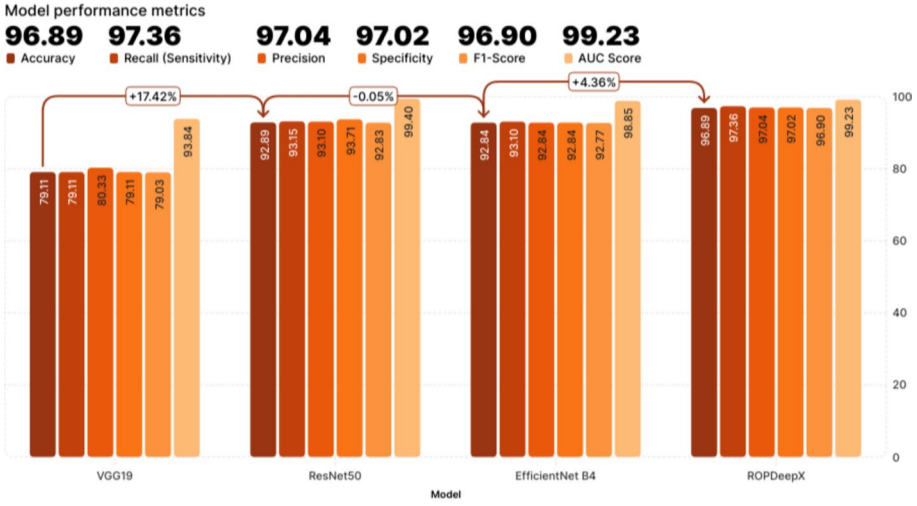


Fig. 9 ROPDeepX demonstrates superior performance across all metrics against VGG19, ResNet50, and EfficientNet-B4 using FARFUM-RoP Dataset

4.4 Multi-class classification performance

The results of the multi-class classification evaluation for ROPDeepX on the FARFUM-RoP and RIDIRP datasets are impressive (table 6), particularly in terms of accuracy and model performance across all three classes: Normal, Pre-Plus, and Plus.

- Accuracy: ROPDeepX achieved an outstanding accuracy of 99.7% on the RIDIRP dataset, showcasing its exceptional ability to classify retinal images with a high degree of correctness. The model demonstrated consistent accuracy across both datasets, further emphasizing its reliability.
- Precision, recall, and F1-score confirmed the model's balanced performance across all categories, indicating that ROPDeepX maintains consistent accuracy without bias toward any specific class, thereby ensuring reliable performance in real-world applications
- Confusion Matrix: Analysis of the confusion matrix Fig. 10 revealed that ROPDeepX performs particularly well in detecting Plus disease, a critical stage in Retinopathy of Prematurity (ROP) diagnosis. The model displayed high sensitivity for this class, ensuring that Plus disease is effectively identified, which is crucial for early intervention. Moreover, ROPDeepX demonstrated robust discrimination between the Pre-Plus and Plus stages, which are often visually subtle and challenging to differentiate in medical imaging. Confusion matrices provide critical insights beyond accuracy, revealing detailed error patterns and supporting a deeper evaluation of model behavior in multi-class medical classification tasks [46].

These results indicate that ROPDeepX not only provides a high level of precision in multiclass classification but also contributes to a more reliable and scalable solution for early ROP detection. By enhancing the diagnostic capabilities in identifying the different stages of ROP, the model holds great potential for clinical use, where early intervention can significantly reduce the risk of vision impairment in premature infants. The success of ROPDeepX in multiclass classification exemplifies the growing potential of deep learning in improving healthcare outcomes through automated, accurate disease diagnosis.

Table 6 Multi-class ROP classification performance on two retinal image datasets

Dataset	Metric	VGG19 (%)	ResNet50 (%)	EfficientNet-B4 (%)	ROPDeepX (%)
FARFUM-ROP (3-class)	Accuracy	79.1	92.9	92.8	**96.9**
	Recall (Sens.)	79.1	93.1	93.1	**97.4**
	Precision	80.3	93.1	92.8	**97.0**
	Specificity	79.1	93.7	92.8	**97.0**
	F1-Score	79.0	92.8	92.8	**96.9**
	AUC	93.8	98.4	98.9	**99.2**
RIDIRP (3-class)	Accuracy	89.8	85.8	89.9	**99.7**
	Recall (Sens.)	69.0	78.2	89.4	**99.7**
	Precision	86.2	88.6	91.7	**99.7**
	Specificity	78.0	75.3	85.0	**99.8**
	F1-Score	86.5	87.0	90.4	**99.7**
	AUC	94.8	95.4	95.6	**100.0**

Bold values indicates the best

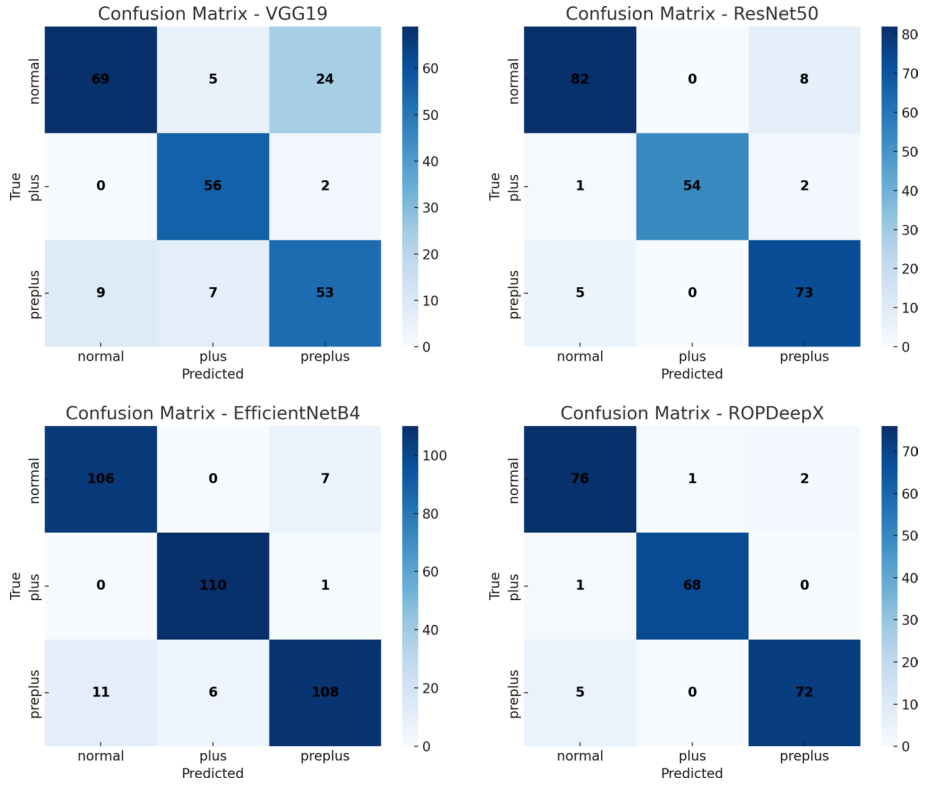


Fig. 10 Confusion matrix comparison of four CNN architectures –VGG19, ResNet50, EfficientNet-B4, and the proposed ROPDeepX

4.5 Binary classification performance

Building on the multiclass classification results, where ROPDeepX demonstrated robust performance in distinguishing Normal, Pre-Plus, and Plus stages, it is equally important to evaluate the model in a binary classification setting. Binary classification–differentiating simply between Normal and ROP conditions–provides a clearer perspective on the model's ability to detect the presence of disease. This analysis complements the multiclass findings by focusing on overall disease detection sensitivity and specificity, which are critical for clinical screening applications.

Across the three binary classification datasets in Table 7, ROPDeepX consistently outperformed VGG19, ResNet50, and EfficientNet-B4 in nearly all metrics, demonstrating its robustness and generalization capability. On FARFUM-ROP, ROPDeepX achieved the highest scores across accuracy, recall, precision, specificity, F1-score, and AUC, with notable gains over other models–particularly in precision and recall–indicating a strong ability to correctly identify both positive and negative cases. For RIDIRP, ROPDeepX maintained superior performance, especially in accuracy and recall, suggesting its effectiveness in handling challenging inter-class variations. Even on the more difficult HVDROPDB v3 dataset, where all models experienced performance drops, ROPDeepX led in every metric, achieving a substantial margin in accuracy and precision compared to others. These results highlight that ROPDeepX not only achieves the highest classification accuracy but also maintains balanced sensitivity and specificity, making it the most reliable model across diverse binary classification scenarios.

Receiver Operating Characteristic (ROC) curves for four convolutional neural network models–VGG19, ResNet50, EfficientNet-B4, and ROPDeepX applied to three

Table 7 Binary ROP classification performance on three datasets

Dataset	Metric	VGG19 (%)	ResNet50 (%)	EfficientNet-B4 (%)	ROPDeepX (%)
FARFUM-ROP (2-class)	Accuracy	93.4	94.6	95.1	**98.0**
	Recall (Sens.)	96.5	97.1	95.9	**98.0**
	Precision	85.4	85.7	90.7	**98.0**
	Specificity	91.9	92.5	94.7	**97.3**
	F1-Score	90.6	91.9	93.2	**98.0**
	AUC	97.6	98.6	98.6	**99.8**
RIDIRP (2-class)	Accuracy	83.3	87.6	90.4	**92.4**
	Recall (Sens.)	83.9	88.9	90.2	**92.4**
	Precision	87.8	89.7	89.6	**92.5**
	Specificity	82.4	85.9	90.6	**92.4**
	F1-Score	85.8	89.3	89.9	**92.4**
	AUC	86.5	89.6	94.2	**94.8**
HVDROPDB v3 (2-class)	Accuracy	80.0	80.2	80.0	**93.3**
	Recall (Sens.)	77.8	87.5	75.0	**93.8**
	Precision	87.5	77.8	85.7	**94.2**
	Specificity	83.3	71.4	85.7	**93.8**
	F1-Score	82.4	82.4	80.0	**93.3**
	AUC	94.4	87.5	80.4	**94.6**

Bold values indicates the best

binary ROP classification tasks (Normal vs ROP) across the FARFUM-RoP, RIDIR, and HVDROPDB V3 datasets. The ROC curve evaluates a model's sensitivity versus its false positive rate, with curves closer to the top-left corner indicating better classification performance. ROPDeepX demonstrates superior area under the curve (AUC) across all datasets, reflecting its enhanced discriminatory power in detecting ROP. ROC analysis provides a robust assessment of classifier behavior independent of class imbalance or decision thresholds, making it a crucial metric in medical AI evaluation (See Fig. 11).

These findings confirm that ROPDeepX delivers consistent and superior binary classification performance across datasets of varying difficulty and acquisition quality. Its combination of high sensitivity and specificity ensures both effective disease detection and reduction of false alarms, making it well-suited for real-world clinical screening workflows

Overall,Fig. 12 shows a comparison of four CNN architectures (VGG19, ResNet50, EfficientNet-B4, and ROPDeepX) across five datasets (FARFUM 3-class, RIDIRP 3-class, FARFUM 2-class, RIDIRP 2-class, HVDROPDB 2-class) using six evaluation metrics: Accuracy, Sensitivity, Precision, Specificity, F1-Score, and AUC. ROPDeepX consistently outperforms baseline models, particularly in binary classification tasks, demonstrating its robustness and discriminative power in identifying Retinopathy of Prematurity.

ROPDeepX maintains top-tier performance in both multi-class and binary scenarios, suggesting robust generalization. In contrast, the baseline models (VGG19, ResNet50, EfficientNet-B4) show more varied performance and generally struggle to match ROPDeepX, especially on the more complex 3-class classification problems.

4.6 Dataset adoption and reproducibility

The release of the FARFUM-RoP, RIDIRP, and HVDROPDB datasets in 2024 has greatly enhanced the transparency and comparability of ROP research. Earlier studies often relied on proprietary or single-center image collections, which made replication difficult and limited opportunities for direct performance comparison. The studies summarized

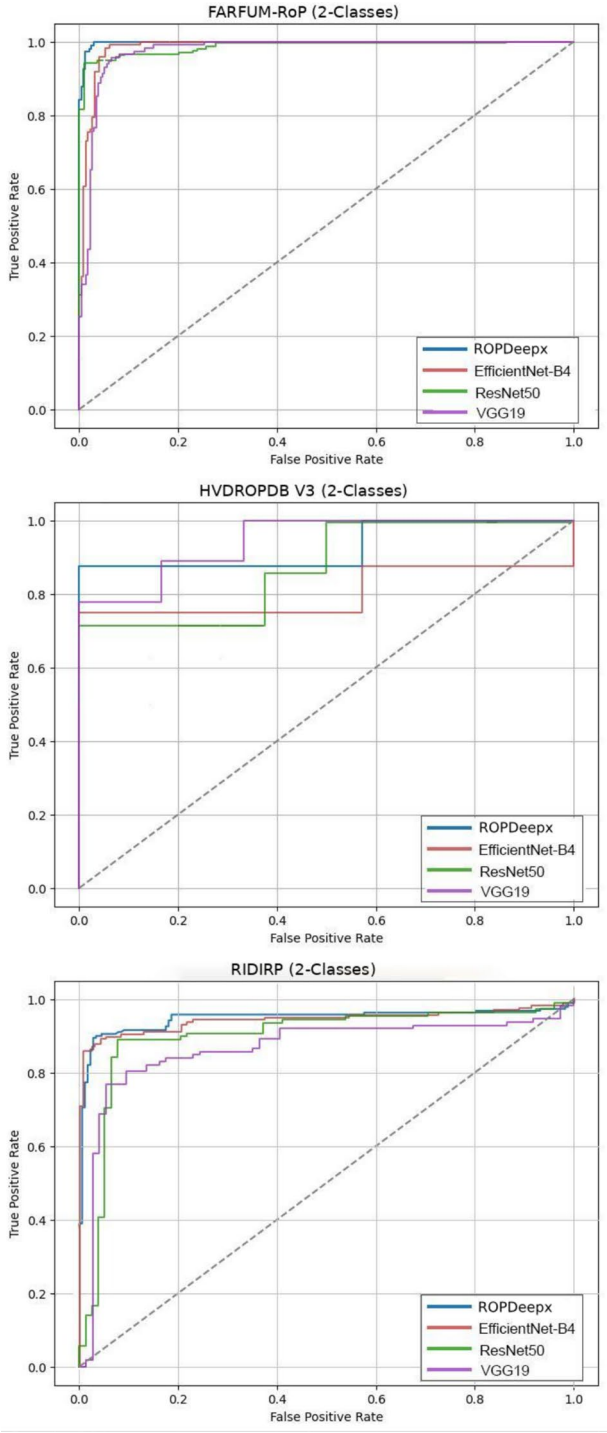
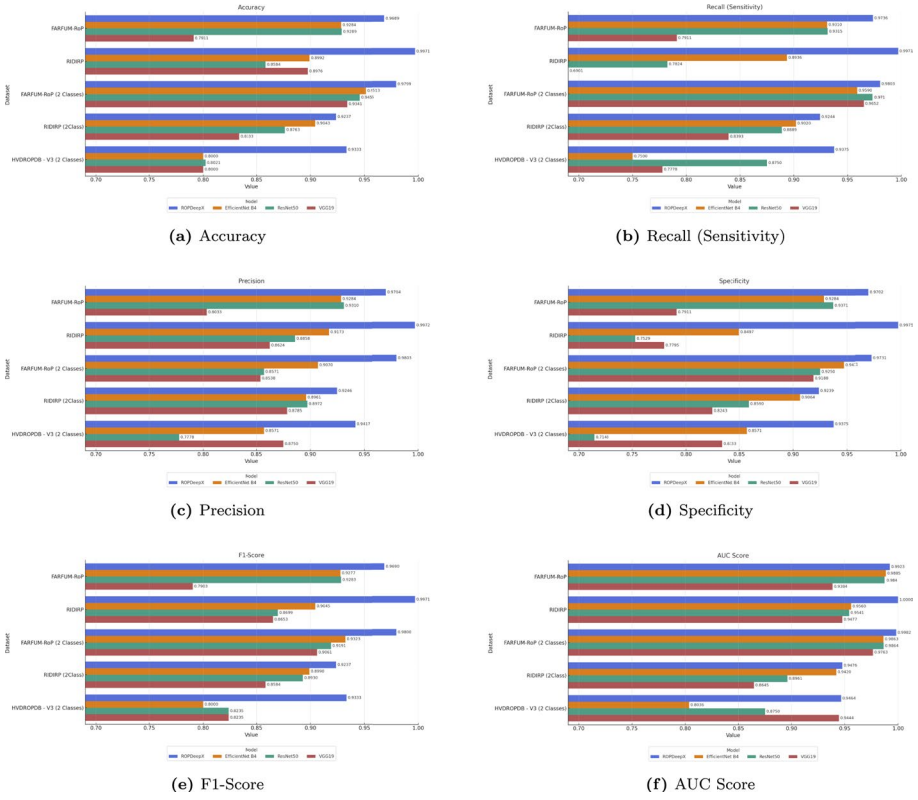


Fig. 11 ROC curve comparison of CNN architectures for binary ROP classification on three datasets

in Table 8 illustrate a growing trend toward leveraging these public datasets to promote reproducibility and cross-study benchmarking. One example is the development of a hybrid deep learning model specifically trained and evaluated using only open-access datasets, enabling fair comparisons and reliable generalization testing across different populations. Collectively, this trend reflects a broader community shift toward collaborative data sharing in neonatal ophthalmology.



(a) Accuracy (b) Recall (Sensitivity)

(c) Precision (d) Specificity

(e) F1-Score (f) AUC Score

Fig. 12 Comparative evaluation of ROPDeepX and baseline models (VGG19, ResNet50, EfficientNet-B4) across six metrics on multiple ROP datasets. The figure includes Accuracy, Recall (Sensitivity), Precision, Specificity, AUC Score, and F1-Score

ROPDeepX advances this trend by being the first to incorporate all three major open datasets into a unified evaluation framework. The model was trained exclusively on FARFUM-RoP and tested on completely unseen datasets – RIDIRP and HVDROPDB – to evaluate cross-dataset generalization. This approach directly addresses a long-standing limitation in medical AI: the tendency for models to perform well on their development dataset but fail to maintain accuracy when applied to external populations or different imaging devices.

4.7 Error analysis and interpretability

To better understand model limitations, a detailed error analysis was conducted using the misclassified samples across all folds. Most errors occurred at the Pre-Plus and Plus boundary, where retinal vessel tortuosity and dilation appear moderate or transitional. These borderline cases often exhibit subtle features that even expert ophthalmologists find challenging to classify consistently. Grad-CAM heatmaps (Fig. 13) reveal that both the ResNet-50 and EfficientNet-B4 branches focused on clinically relevant regions, including the posterior pole and major vascular arcades, confirming that the model's attention mechanisms align with anatomical features used by human graders. However, differences in illumination, image contrast, and peripheral artifacts occasionally led to incorrect predictions. Importantly, these misclassifications were not random; they clustered around ambiguous disease stages rather than normal eyes, demonstrating that ROPDeepX errs within clinically plausible margins. The overall consistency of

Table 8 Summary of recent ROP detection and classification studies with datasets, models, and performance metrics

Title	Dataset	Model	Key performance metrics
Automated Diagnosis of Plus Disease in ROP based on Transformer-based Unsupervised Curriculum Learning (2025) [47]	Part of FARFUM-RoP (76 posterior pole retinal images)	Vision Transformer with unsupervised curriculum learning to detect Plus disease	Sensitivity: 94–98%, Specificity: 94–99% (Plus vs. non-Plus)
Enhancing ROP Plus Form Diagnosis: An Automatic Blood Vessel Segmentation Approach for Newborn Fundus Images (2024) [48]	RIDIRP	Hybrid image-processing segmentation of blood vessels for Plus disease assessment	Vessel segmentation specificity: ∼97.4%, Sensitivity: ∼83–84% (on benchmarks)
Classification of Stages 1, 2, 3 and Preplus, Plus Disease of ROP using MultiCNN_LSTM Classifier (2025) [13]	HVDROPDB	MultiCNN_LSTM combining multiple CNN feature extractors with LSTM for stage and Plus classification	Stage classification accuracy: ∼90%; Plus classification (3-class): 85% accuracy
Deep Learning Framework for Classification and Explanation of Early ROP Stages from HVDROPDB Dataset (2024) [49]	HVDROPDB	Deep CNN with Explainable AI (Grad-CAM) for early-stage ROP classification	Early-stage classification accuracy: ∼88–90%, with visual explanations via heatmaps
Retinopathy of Prematurity Disease Diagnosis using Deep Learning (2024) [18]	HVDROPDB	ResNet50 combined with Deep Neural Network (DNN) for diagnosis	Accuracy: 92.8%, Sensitivity: 94.9% (Kaggle + HVDROPDB datasets)
The Proposed Model	FARFUM-RoP, RIDIRP, HVDROPDB	Hybrid CNN combining ResNet50 + EfficientNet-B4 with soft attention fusion; trained on FARFUM-RoP and tested on RIDIRP & HVDROPDB	Multiclass accuracy: up to 99.7%; Binary classification AUC: up to 99.8%

Grad-CAM activations and the low variance of fold-wise results support the interpretability, robustness, and reliability of the proposed hybrid network.

the Panels in Fig. 13 show two representative errors with heatmaps from the ResNet-50 branch (left in each pair) and the EfficientNet-B4 branch (right). Both backbones attend to clinically meaningful regions–posterior pole and vascular arcades–despite the incorrect predictions, illustrating that most mistakes arise from borderline Pre-Plus and Plus presentations rather than off-target attention.

4.8 Limitations

Despite these encouraging results, several factors limit the immediate clinical adoption of ROPDeepX. First, the evaluation was conducted solely on curated public datasets, and the model's performance in prospective, real-time screening environments has yet to be validated. Such settings introduce challenges like inconsistent image quality, operator variability, and patient movement, which could impact accuracy. Second, although the datasets span multiple countries, the geographic and demographic diversity remains limited, potentially affecting generalizability to underrepresented populations. Another limitation is the model's reliance on pre-trained networks. While transfer learning helps reduce training time and memory requirements, the choice of pre-trained weights (e.g., from ImageNet) may not always be optimal for medical image tasks. Fine-tuning these pre-trained models on specialized medical datasets could further enhance performance, especially in distinguishing subtle features specific to ROP.

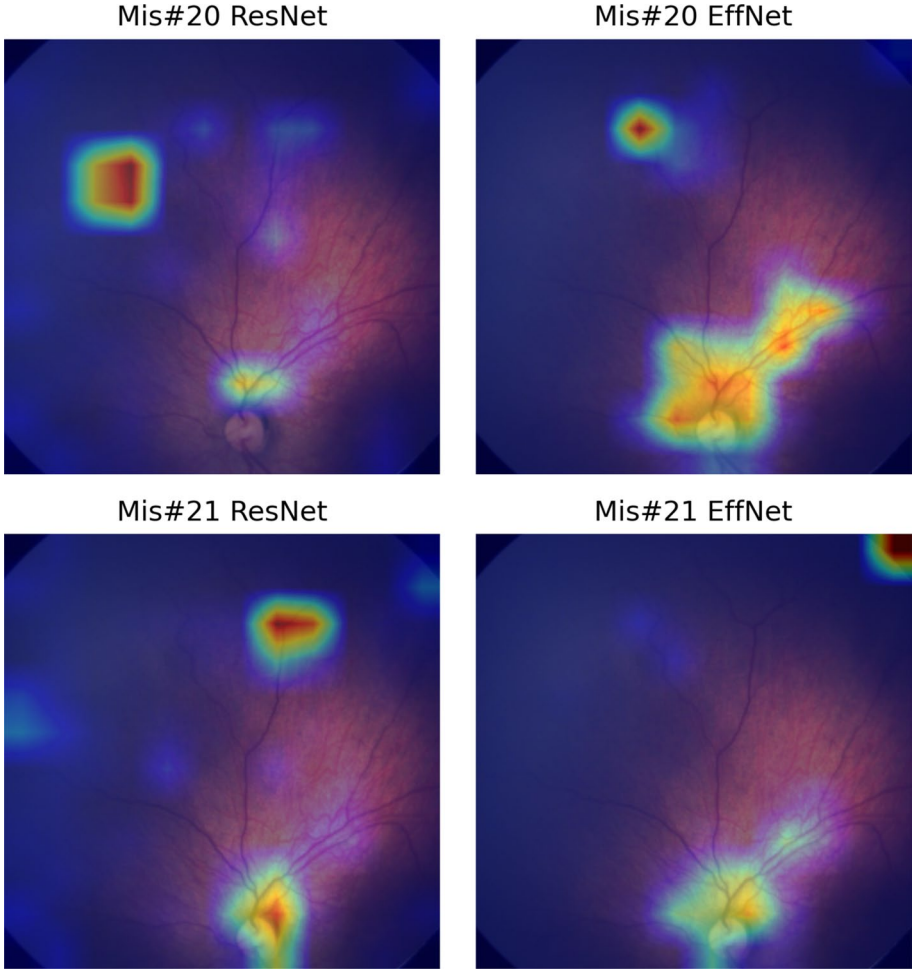


Fig. 13 Grad-CAM overlays for misclassified test images

4.9 Future work

Future research should aim to deploy ROPDeepX in live neonatal screening programs, capturing performance metrics in real-world clinical contexts. Expanding dataset diversity–especially from low-resource settings and varied imaging devices–will help ensure broader applicability. Further, integrating advanced explainability mechanisms and quantifiable attention maps can enhance clinician trust and model transparency. Finally, developing an end-to-end deployment pipeline that integrates preprocessing, inference, and reporting into existing hospital IT systems will be crucial for translating research performance into clinical impact.

5 Conclusion

ROPDeepX represents a significant advancement by integrating multiple open datasets, an innovative hybrid CNN architecture, and extensive evaluation protocols into a single, cohesive study. This integration addresses key limitations in earlier research – particularly the lack of cross-dataset validation and restricted methodological scope.

By exclusively using publicly available datasets, ROPDeepX ensures transparency, reproducibility, and fair benchmarking, while its high cross-dataset performance highlights its clinical applicability in real-world settings. The model's calibration and

reliability, as demonstrated through metrics like ECE and Brier Score, further enhance its clinical trustworthiness. By training on FARFUM-RoP and validating on RIDIRP and HVDROPDB, ROPDeepX achieves state-of-the-art performance while demonstrating strong generalization to unseen data. The hybrid architecture's superior feature representation and the model's consistent results across diverse datasets suggest its potential for real-world deployment in neonatal screening programs.

The findings affirm that multi-dataset training and rigorous cross-dataset testing are essential for developing clinically robust ROP detection systems. ROPDeepX's methodology not only sets a new benchmark for performance but also establishes a blueprint for future research: leveraging the full breadth of available public datasets, integrating complementary model architectures, and evaluating with a comprehensive set of clinically relevant metrics. This approach aligns with the broader goal of creating AI systems that can be trusted across different clinical settings, ultimately contributing to the early detection and prevention of ROP-related blindness worldwide.Future work should focus on validating ROP Models in real-world clinical settings, assessing real-time inference performance, and exploring its integration into neonatal screening pipelines to bridge the gap between research and clinical adoption.

Author contributions
Esraa Mohiy: Methodology, Software, Resources, Data curation, Visualization, Writing—review & editing. Hamada I. AbdulWakel: Methodology, Investigation, Conceptualization, Writing—review and editing. Marwa Khairy: Methodology, Validation, Conceptualization, Writing—review and editing. Essam H. Houssein: Supervision, Methodology, Formal analysis, Writing—review andediting. All authors read and approved the final paper.

Funding
This research did not receive any specific grant or funding.

Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
This article does not contain any studies with human participants or animals carried out by any of the authors.

Consent for publications
Not applicable

Clinical trial registry
Not applicable

Competing interests
The authors declare no Conflict of interest.

Received: 15 August 2025 / Accepted: 1 January 2026
Published online: 26 March 2026

References

1. Şahin S, Özdemir MA, Çetin EN, Küçüktaşçı K, Ergin H. Can the side effects of mydriatics be reduced with the ROP-bundle protocol? Namık Kemal Med J. 2023;11:271–5.
2. Jang JH, Kim YC. Retinal vascular development in an immature retina at 33–34 weeks postmenstrual age predicts retinopathy of prematurity. Sci Rep. 2020;10:18111.
3. Fierson WM, Chiang MF, Good W, Phelps D, Reynolds J, Robbins SL, et al. Screening examination of premature infants for retinopathy of prematurity. Pediatrics. 2018;142(6):e20183061.
4. Hong EH, Shin YU, Cho H. Retinopathy of prematurity: a review of epidemiology and current treatment strategies. Clin Exp Pediatr. 2021;65(3):115.
5. Kaur K, Mikes BA. Retinopathy of Prematurity. StatPearls Publishing, Treasure Island (FL), 2025. [Updated 2025 Jun 2].
6. Solarte CE, Awad AH, Wilson CM, Ells A. Plus disease: Why is it important in retinopathy of prematurity? Middle East African J Ophthalmol. 2010;17(2):148–55.
7. Huang K, Dong W, Li J, Chen Y, Zhong J, Yi Z. GFF-Net: graph-based feature fusion network for diagnosing plus disease in retinopathy of prematurity. Appl Intell. 2023;53(21):25259–81.

8. Jafarizadeh A, Maleki SF, Pouya P, Sobhi N, Abdollahi M, Pedrammehr S, et al. Current and future roles of artificial intelligence in retinopathy of prematurity. Artif Intell Rev. 2025;58(6):1–55.
9. Ells AL, Gole GA, Lloyd Hildebrand P, Ingram A, Wilson CM, Williams RG. Posterior to the ridge laser treatment for severe stage 3 retinopathy of prematurity. Eye (London). 2013;27(4):525–30.
10. Akbari M, Pourreza HR, Khalili Pour E, Dastjani Farahani A, Bazvand F, Ebrahimiadib N, et al. FARFUM-RoP: a dataset for computer-aided detection of retinopathy of prematurity. Scientific Data. 2024;11(1):1176.
11. Timkovič J, Nowaková J, Kubíček J, Hasal M, Varyšová A, Kolarčík L, et al. Retinal image dataset of infants and retinopathy of prematurity. Sci Data. 2024;11(1):814.
12. Agrawal R, Walambe R, Kotecha K, Gaikwad A, Deshpande CM, Kulkarni S. Hvdropdb datasets for research in retinopathy of prematurity. Data Brief. 2024;52:109839.
13. Agrawal R, Kulkarni S, Deshpande M, Gaikwad A, Walambe R, Kotecha KV. Classification of stages 1, 2, 3 and preplus, plus disease of ROP using multiCNN_LSTM classifier. MethodsX. 2025;14:103182.
14. Huang P, Xie Y, Rong W, Lin Q, Cai N, Chen H, et al. Roprnet: deep learning-assisted recurrence prediction for retinopathy of prematurity. Biomed Signal Process Control. 2025;100:107135.
15. Salih N, Ksantini M, Hussein N, Ben Halima D, Abdul Razzaq A, Ahmed S. Prediction of ROP zones using deep learning algorithms and voting classifier technique. Int J Comput Intell Syst. 2023;16(1):86.
16. Ramanathan A, Athikarisamy SE, Lam GC. Artificial intelligence for the diagnosis of retinopathy of prematurity: a systematic review of current algorithms. Eye. 2023;37(12):2518–26.
17. Sankari VR, Umapathy S. Computer-aided diagnosis of early-stage retinopathy of prematurity in neonatal fundus images using artificial intelligence. Biomed Phys Eng Express. 2024;11(1):015015.
18. Ndunge Mutua E, Shibwabo Kasamani B, Reich C. Retinopathy of prematurity disease diagnosis using deep learning. Int J Comput Digital Syst. 2024;16(1):1097–110.
19. Matamala Avrova E, Camara O, Valls A. A deep learning approach for the segmentation and classification of plus disease in retinal fundus images of preterm infants. J Clin Med. 2024;12(7):1607.
20. Tong Y, Wei L, Deng Q, Chen C, Shen Y. Automated identification of retinopathy of prematurity by image-based deep learning. Eye Vis. 2020;7(1):40.
21. Chen JS, Coyner AS, Ostmo S, Sonmez K, Bajimaya S, Pradhan E, et al. Deep learning for the diagnosis of stage in retinopathy of prematurity: accuracy and generalizability across populations and cameras. Ophthalmol Retina. 2021;5(10):1027–35.
22. Feng W, Huang Q, Ma T, Ju L, Ge Z, Chen Y, et al. Development and validation of a semi-supervised deep learning model for automatic retinopathy of prematurity staging. Iscience. 2024;27(1):108516.
23. Attallah O. DIAROP: automated deep learning-based diagnostic tool for retinopathy of prematurity. Diagnostics. 2021;11(11):2034.
24. Mulay S, Ram K, Sivaprakasam M, Vinekar A (2019) Early detection of retinopathy of prematurity stage using deep learning approach. In: Medical Imaging 2019: Computer-Aided Diagnosis, SPIE. vol 10950, pp 758–764
25. Tan Z, Simkin S, Lai C, Dai S. Deep learning algorithm for automated diagnosis of retinopathy of prematurity plus disease. Transl Vis Sci Technol. 2019;8(6):23–23.
26. Ryan MC et al (2014) Development and evaluation of reference standards for image-based telemedicine diagnosis and clinical research studies in ophthalmology. In AMIA Annual Symposium Proceedings, American Medical Informatics Association, vol 2014, p 1902
27. Imaging and informatics in retinopathy of prematurity. https://irop.github.io/index.html. Accessed 16 Sept 2021.
28. Ataer-Cansizoglu E, et al. Computer-based image analysis for plus disease diagnosis in retinopathy of prematurity: performance of the i-ROP system and image features associated with expert diagnosis. Transl Vis Sci Technol. 2015;4(2):5–5.
29. Worrall DE, Wilson CM, Brostow GJ (2016) Automated retinopathy of prematurity case detection with convolutional neural networks. In Deep Learning and Data Labeling for Medical Applications, Springer. pp 68–76
30. Yildiz VM, et al. Plus disease in retinopathy of prematurity: convolutional neural network performance using a combined neural network and feature extraction approach. Transl Vis Sci Technol. 2020;9(2):10–10.
31. Ding A, Chen Q, Cao Y, Liu B (2020) Retinopathy of prematurity stage diagnosis using object segmentation and convolutional neural networks. In: 2020 International Joint Conference on Neural Networks (IJCNN), IEEE, pp 1–6
32. Walz JM, et al. The german rop registry: data from 90 infants treated for retinopathy of prematurity. Acta Ophthalmol. 2016;94(7):e744–52.
33. Hong EH, et al. Nationwide incidence and treatment pattern of retinopathy of prematurity in south Korea using the 2007–2018 national health insurance claims data. Sci Rep. 2021;11(1):1–10.
34. Wang J, et al. Automated retinopathy of prematurity screening using deep neural networks. EBioMedicine. 2018;35:361–8.
35. Chiang MF, Wang L, Busuioc M, Du YE, Chan P, Kane SA, et al. Telemedical retinopathy of prematurity diagnosis: accuracy, reliability, and image quality. Arch Ophthalmol. 2007;125(11):1531–8.
36. Tian P et al (2019) A severity score for retinopathy of prematurity. In: Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining, ACM, pp 1809–1819
37. Zhang Y. Dnn classifier of wide-angle retinal images in computer-aided screening for ROP. 2018. https://doi.org/10.6084/m9.figshare.6367685.v1.
38. Huang YP, Vadloori S, Chu HC, Kang EYC, Wu WC, Kusaka S, et al. Deep learning models for automated diagnosis of retinopathy of prematurity in preterm infants. Electronics. 2020;9(9):1444.
39. Bhatkalkar B, Joshi A, Prabhu S, Bhandary S. Automated fundus image quality assessment and segmentation of optic disc using convolutional neural networks. Int J Electr Comput Eng (2088-8708). 2020;10(1):816–27.
40. Murickan T, Coyner AS, Ostmo S, Moshfeghi DM, Chan RVP, Kalpathy-Cramer J, et al. External validation of autonomous retinopathy of prematurity screening in the sundrop program. Investig Ophthalmol Visual Sci. 2023;64(8):268–268.
41. Wang SK, Callaway NF, Wallenstein MB, Henderson MT, Leng T, Moshfeghi DM. Sundrop: six years of screening for retinopathy of prematurity with telemedicine. Canadian J Ophthalmol. 2015;50(2):101–6.
42. Quinn GE, Ying GS, Daniel E, Hildebrand PL, Ells A, Baumritter A, Kemper AR, Schron EB, Wade K, e ROP Cooperative Group, et al (2014) Validity of a telemedicine system for the evaluation of acute-phase retinopathy of prematurity. JAMA Ophthalmol. 132(10):1178–84
43. Wade KC, Ying GS, Baumritter A, Gong A, Kemper AR, Quinn GE, e ROP Cooperative Group, et al. Factors in premature infants associated with low risk of developing retinopathy of prematurity. JAMA Ophthalmol. 2019;137(2):160–6.

44. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 770–778
45. Tan M, Le Q (2019) Efficientnet: rethinking model scaling for convolutional neural networks. In International conference on machine learning, PMLR. pp 6105–6114
46. Sokolova M, Lapalme G. A systematic analysis of performance measures for classification tasks. Inf Process Manag. 2009;45(4):427–37.
47. Deepthi K, Josephine MS, Raja VJ. Automated diagnosis of plus disease in retinopathy of prematurity based on transformer-based unsupervised curriculum learning. Biomed Signal Process Control. 2025;104:107521.
48. Almeida J, Kubicek J, Penhaker M, Cerny M, Augustynek M, Varysova A, et al. Enhancing ROP plus form diagnosis: an automatic blood vessel segmentation approach for newborn fundus images. Res Eng. 2024;24:103054.
49. Agrawal R, Kulkarni S, Gaikawad A, Deshpande CM, Kotecha K, Walambe R (2024) Deep learning framework for classification and explanation of early rop stages from hvdropdb dataset. In 2024 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC), IEEE, pp 1–6

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.